

Conjecture 00000001255 is false

A noisy one-site cellular automaton

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Abstract

A binary flip rule on a one-site periodic ring has no deterministic fixed configuration. Its nondegenerate noisy update has a unique stationary law, the uniform law. This law is an extreme stationary measure and is not a point mass. Thus the conjecture's extreme-point assertion fails.

1 Statement and model

The conjecture asserts, for a cellular automaton on a finite ring driven by Glauber-type noise, that the relevant stationary simplex has deterministic fixed points as its extreme points. It also asserts a support description and a dimension identity. A counterexample to the extreme-point clause suffices to refute their conjunction.

Take the periodic ring $\mathbb{Z}/\mathbb{Z}$, with binary state space $S = \{0, 1\}$. The local deterministic rule is $F(s) = 1 - s$. At an update, independently flip the prescribed output with probability $1/4$. Equivalently, apply F with probability $3/4$, and keep the current bit with probability $1/4$. This is a one-site, positive-temperature Glauber-type noisy local update. In the order $(0, 1)$ its transition matrix is

$$P = \begin{pmatrix} 1/4 & 3/4 \\ 3/4 & 1/4 \end{pmatrix}.$$

Every entry is positive and each row sums to one. The deterministic rule has a two-cycle and no fixed point.

2 Exact stationary and extreme-point computation

Theorem 1. *The only stationary probability law is $u = (1/2, 1/2)$. It is an extreme point of the convex set of stationary laws, and is not a Dirac law.*

Proof. Write a probability law as (a, b) , with $a, b \geq 0$ and $a+b = 1$. Stationarity at state zero gives $a = a/4 + 3b/4$, hence $a = b = 1/2$. Conversely, direct multiplication gives $uP = u$. The stationary-law set is therefore the singleton $\{u\}$. If $u = rv + (1-r)w$ with stationary v, w and $0 < r < 1$, uniqueness implies $v = w = u$, which is exactly the extreme-point condition. Both coordinates of u are $1/2$, so it is neither of the two Dirac laws. $\square$

The unique extreme stationary law cannot be supported at a deterministic fixed point because there are none. If the statement instead intends the simplex of probability laws on the stationary support, that support is all of S ; its vertices are the two Dirac laws, and neither underlying configuration is a fixed point of F . Thus that reading fails as well. No interpretation of the undefined “minimal attractor dimension” is needed for this disproof.

3 Formal verification

The accompanying Lean project represents the two configurations by `Bool`, and probability laws by functions $S \rightarrow \mathbb{R}$ with nonnegative coordinates summing to one. This represents every probability measure on the finite state space, not a restricted numerical sample. The positivity and row-sum theorems verify the kernel. `stationary_unique` quantifies over all real-valued laws; `uniform_extreme` uses the standard convex-combination definition. `conjecture_false` negates the extreme-point clause, using the actual flip rule and the actual stationary set.

Lean and Mathlib are pinned by the toolchain and Lake manifest. Run `lake build` and `lake env lean` `Check.lean` in the `lean` directory. No admitted proofs, additional axioms, or `native_decide` are used. The axiom audit is in `verification.txt`. Rebuild this document with `pdflatex main.tex` twice.